

Buff Stooges

Software Requirements Specifications for:

Buff Stooges Project

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1. Introduction

1.1. Purpose of the document

The purpose of this document is to provide a comprehensive set of requirements for the Buff Stooges fitness workout. It outlines the functional and non-functional requirements for the software system, the design models that were derived from them, and the constraints and assumptions that impact the development process.

This document serves as a communication tool between The Two Stooges development team (Bradley and Jakub) and Professor Mohsen Saffari. It provides a clear understanding of the requirements for the software system, enabling stakeholders to make informed decisions about the design, development, testing, and deployment of the system.

The requirements outlined in this document are intended to guide the development team in the creation of a high-quality software system that meets the demands of PNW ECE 35400. The document also serves as a baseline for testing and verification, ensuring that the software system meets the desired functionality and quality standards. To this end, the IDs of relevant functional and non-functional requirements from section 3 are referenced throughout this document in the form (FR#) and (NFR#), respectively.

1.2. Scope of the project

The Buff Stooges fitness workout is an offline software that enables users to plan workouts, record training, and track nutrition and progress. The scope of this project is to develop a high-quality software system that provides a streamlined and intuitive user experience for planning workouts, tracking macros, and managing user data. The system is a Java desktop application for Windows platforms.

The software system will include the following key features:

- **Profiles:** Multiple local user profiles with a preferred unit system (FR1).
- **Exercises and plans**: A built-in exercise library searchable by muscle group, and reusable named workout plans (FR2, FR3).
- **Workout tracking:** Workout sessions with logged sets and cardio time, a rest timer, and hydration reminders (FR4, FR5, FR6).
- **Nutrition and body metrics:** BMI with category, a food and macro log, a daily calorie target with warnings, and a recommended protein range (FR7, FR8, FR9, FR10).
- **Progress tracking:** A muscle group list showing when each was last exercised, a weekly workout streak, and a 12-month activity calendar (FR11, FR12, FR13).
- **Dashboard:** A home screen that summarizes today's macros, calories versus target, and the current streak (FR14).

- **Data handling**: Input validation and automatic saving of each user's data to a JSON file (FR15, FR16).

The following items are outside the scope of this project:

- Network features of any kind (NFR7).
- Platforms other than Windows (e.g. mobile and web)

The development of the software system has so far followed the phases of this course: requirements analysis (Phase 1), class design (Phase 2.1), scenario and sequence modeling (Phase 2.2), and this SRS specification document (Phase 2.3). Following this phase is the implementation across four versions and with good engineering practice. The project timeline is approximately 1½ months, with the project submission date being December 5th, 2026.

In summary, the scope of this project is to develop a focused, offline Windows application that helps beginners learn what a balanced routine looks like and help avid gym-goers plan, record, and review one's training and nutrition in one place.

1.3. Stakeholders involved

The following stakeholder groups have an interest in the Buff Stooges workout app. The app is intended for users of all experience levels, from beginners to avid gym-goers (those two groups are the primary stakeholders). Influencers, gym owners, and the development team are secondary.

1.3.1. Beginners

Wanting to get started but not sure where to start. They benefit most from the searchable exercise library (FR2), reusable plans (FR3), the BMI category (FR7), the protein recommendation (FR10), and the explanation of streaks shown to new users (FR12). The first-use usability requirement (NFR8) exists for them.

1.3.2. Avid Gym-Goers

This group includes bodybuilders, physique-focused athletes, and cardio-focused athletes. They want detailed session logging (FR4), calorie and macro tracking with targets (FR8, FR9, FR10), a view of which muscles are due for training (FR11), and motivation tools such as the weekly streak and activity calendar (FR12, FR13).

1.3.3. Influencers

Influencers include social media personalities, medical recommenders, and trainers. They may recommend the app to others, so it should follow accepted practices. For that reason, we shall research sound advice on topics such as protein ranges.

1.3.4. Gym Owners

The success of our app means more customers for gym owners. They have no direct interaction with the system, so their needs are served indirectly by making the app useful and approachable for gym members of every level.

1.3.5. The Development Team

We (Bradley Goralczyk and Jakub Niedzielski) are both developers and product owners. The team needs a complete, testable, and traceable specification that can be delivered within the available time to get a good grade for this assignment. This is the greatest stake of all.

1.4. Overview of the Buff Stooges workout app

1.4.1. Problem statement

Physical fitness is a science not commonly understood by those at the start of one's fitness journey. There is more to a workout than just moving one's body: a person must also adequately take in proteins and nutrients, exercise the whole body evenly, and get enough rest for muscles to heal and grow. The Buff Stooges app shall educate beginners in proven methods and best practices, and enable avid gym-goers to pursue goals and maintain capabilities (building muscle, physique, and/or endurance).

1.4.2. Training

Users choose exercises from a built-in library, assemble them into reusable plans, and log each workout session with sets, weights, repetitions, and cardio time. A rest timer and a hydration reminder support the user during a session.

1.4.3. Nutrition and body metrics

Users log food entries with calories and macronutrients, set a daily calorie target, and receive a warning when one's intake is far from that target. The app also calculates BMI and recommends a daily protein range.

1.4.4. Progress and motivation

A muscle group list shows when each group was last exercised, a weekly stream rewards consistency without punishing rest days, and a 12-month calendar grid shows overall activity. The dashboard brings the day's key numbers together.

1.4.5. Data security

The app is entirely offline. Each user's data is saved automatically to its own JSON file on the local machine. The system makes no network connections.

2. Overall Description

2.1. Product perspective

The Buff Stooges workout app will be a standalone software system designed to provide a streamlined fitness tracking experience for gym-goers. The software system will be developed using Java and will be hosted on the Windows platform.

System interfaces: The system shall interact with its environment in four ways:

- **User interface:** A graphical interface used by one person at time, through which the Dashboard and all other screens are presented.
- **File system:** One JSON file per user profile (FR16)
- **Audio output: T**he computer's sound output, used to play the rest timer alert (FR5).
- **Operating system clock:** Used for dates, session times, timers, and reminders (FR1, FR4, FR5, FR6, FR11, FR12, FR13).

2.2. Product functions

The Buff Stooges workout app will provide the following functions for its users.

2.2.1. Profile management

Create, select, edit, and delete user profiles with name, sex, height, weight, and unit preference, keeping each user's data separate (FR1).

2.2.2. Exercise library and workout planning

Brows and search the built-in exercise library by muscle group, and build reusable named workout plans (FR2, FR3).

2.2.3. Workout tracking and in-session aids

Start and finish workout sessions, log sets and cardio time, and use the rest timer and hydration reminders during a session (FR4, FR5, FR6).

2.2.4. Nutrition and body metrics

Calculate BMI, log food and macros, set a calorie target with warnings, and view a recommended protein range (FR7, FR8, FR9, FR10).

2.2.5. Progress tracking

View muscle groups by when they were last exercised, the weekly streak, and the 12-month activity calendar (FR11, FR12, FR13).

2.2.6. Dashboard

See today's macro totals, calories versus target, and the current streak on the home screen (FR14).

2.2.7. Data validation and storage

Reject invalid numeric input with a clear message and save each user's data automatically in JSON files (FR15, FR16).

2.3. User characteristics

The Buff Stooges workout app has several identified stakeholder groups, but none are more important than those who are working to improve one's one fitness. Whether a user is a beginner or an avid gym-goer, the ideal is that their needs are met. However, there are a few concessions that must be made for the sake of time allotted:

- **Experience:** Beginners may have no fitness knowledge or app experience, so the first-use flow must be learnable without instructions (NFR8). Avid gym-goers expect detailed logging and nutrition tracking.
- **Technical proficiency:** Users are assumed to be comfortable with a basic Windows desktop application.
- **Language:** The Buff Stooges workout app will be available in English only.
- **Units**: Users choose imperial or metric units in one's profile.
- **Shared computers:** Several people may share one computer by using separate profiles. Profiles are not password protected (NFR7).

2.4. Constraints

The development of the Buff Stooges workout app is subject to a number of constraints that must be considered during the design and implementation process. These constraints include:

- **Time constraints:** The project has a tight timeline, with a project submission date of December 5th, 2026.
- **Team constraints:** The system is built by a team of two students who are both developers and product owners.
- **Technology constraints:** The system will be written in Java, run on Windows 10 and 11, and use an existing JSON library for file input and output (NFR1).
- **Offline constraint:** The system shall not use the network (NFR7).
- **Storage constraint:** User data is stored as one JSON file per user (FR16).
- **Budget constraint:** There is no budget, so only free software and libraries are used.

- **Regulatory note:** The app is informational. BMI categories, calorie warnings, and protein ranges are guidance only and are not medical advice.

2.5. Assumptions

- Only one person uses the application at a time.
- Users enter one's own height, weight, and food data honestly and accurately.
- A calendar week runs from Monday to Sunday, and a streak counts consecutive calendar weeks that contain at least one logged workout.
- A daily calorie target of 0 means that no target has been set. No calorie warning is shown in that case.
- Weight of logged sets are stored in kilograms and converted for display when the user prefers imperial units.
- The computer has working sound output for rest timer alert. The visual notification is always shown as well.

2.6. Dependencies

- Java runtime and Windows operating system.
- An existing Java JSON library for reading and writing user files.
- A Java user interface toolkit (JavaFX, Swing, etc.).
- Published nutrition guidance for the protein recommendations (See Appendix 5.3)

2.7. Apportioning of requirements

Because the available time is short, the requirements have been apportioned into priorities. All non-functional requirements apply to the core requirements, so they are not mentioned here.

Priority	Requirements	Rationale
Must Have	FR1, FR2, FR4, FR8, FR9, FR12, FR14, FR15, FR16	Needed for the four user stories, the data handling that supports them, the dashboard, and to keep data valid and saved.
Should Have	FR3, FR5, FR6, FR7, FR10, FR11, FR13	Valuable additions that are independent of the core stories; implemented as time allows.
Could / Won't Have	Data export/import, custom exercises, per-exercise progress charts, settings screen, malformed-file recovery	Not needed for the first release. May be added by the v1.0 revision?

3. Specific Requirements

3.1. Functional Requirements

The system shall...

3.1.1. FR1 – User Profiles

Let users create, select (at startup), edit, and delete profiles (name, sex, height, weight, preferred units: imperial or metric). Each profile data is kept separate.

- Sex is chosen from Male or Female, and preferred units from Imperial or Metric.
- Profiles are stored locally and are not password protected (NFR7).
- A profile owns its workout sessions and food entries, so deleting a profile deletes them as well.

3.1.2. FR2 – Exercise Library

Include a built-in exercise library, tagged by muscle group and equipment, that can be searched by muscle group.

- Each exercise has a name, a list of equipment, a description, and one or more muscle groups.
- The muscle groups are arms, chest, core, glutes, upper legs, lower legs, shoulders, and back.
- A search for a muscle group returns every exercise tagged with it.

3.1.3. FR3 – Plan Builder

Let users create reusable, named workout plans made of an ordered list of exercises.

- Plans are built from exercises in the exercise library (FR2) and can be reused for later sessions.

3.1.4. FR4 – Log workout

Let users start and finish a workout session and log sets (weight, reps) and cardio times and edit or delete logged entries.

- A session has a type (lifting or cardio), a date, a start time, and an end time.
- A logged set records a weight (stored in kilograms) and the number of repetitions.
- Cardio time is a time between the session’s start and end

3.1.5. FR5 – Rest Timer

Provide a rest timer between sets (default 90 s, adjustable) that shows a visual notification and plays a sound at zero.

- The user can change the duration of the timer.

3.1.6. FR6 – Hydration notifier

Show a hydration reminder every 20 minutes during an active workout session.

- A session is active from the moment a user starts it until the user finishes it (FR4).

3.1.7. FR7 – BMI calculator

Calculate and display BMI and its category (underweight, normal, overweight, obese) from height and weight.

- BMI is weight in kilograms divided by height in meters squared. For imperial units, BMI is weight in pounds divided by height in inches squared, multiplied by the constant 703.
- The formula and categories are the same for both sexes.

Underweight BMI	Normal BMI	Overweight BMI	Obese BMI
Below 18.5	18.5 to 25	25 to 30	50 and above

3.1.8. FR8 – Food tracker

Let users log food entries (calories, carbohydrates, protein, fat) per day and edit or delete them.

- Each entry has a name, calories, and grams of carbohydrates, protein, and fat.
- Entries are grouped by date, and a day may contain many entries.
- Negative values are rejected (FR15).

3.1.9. FR9 – Calorie Goal

Let users set a daily calorie target and warn them when the day's logged calories are more than 20% above or below it.

- A target of 0 means that no target is set, and no warning is shown.
- The check runs automatically after each food entry is stored.
- The warning shall not be intrusive as to not interrupt remaining information logging. No pop-up window with confirmation.

3.1.10. FR10 – Protein Goal

Display a recommended daily protein range (grams) calculated from body weight and weekly training volume.

- See Appendix 5.3 for published reference values for protein ranges.

3.1.11. FR11 – Last time exercised

Display the muscle group list with sorting (forward/reverse) and filtering by date last exercised and body category.

- The date last exercised for a muscle group comes from the user's logged sessions.

3.1.12. FR12 – Streak tracker

Track and display the current streak of consecutive calendar weeks containing at least one workout.

- A calendar week with one workout day counts the same as a week with several (because rest days matter).
- Counting stops at the first week that has no logged workout.
- A new user who has no logged workouts is neither streaking nor failing, so instead of a streak number, the dashboard shows a message explaining how streaks work.

3.1.13. FR13 – Calander activity log

Display the past 12 months of workouts as a calendar grid whose color intensity reflects logged activity.

- Each day is one cell. Days with more logged activity are drawn with a more intense color, and days with none use the lightest shade.

3.1.14. FR14 – Main Dashboard

Show a dashboard home screen with today's macro totals, calories vs. target, and the current streak.

- Macro totals are the day's grams of carbohydrates, protein, and fat.
- The dashboard also shows the calorie warning from FR9 when it is active, and the streak explanation from FR12 for new users.

3.1.15. FR15 – Invalid value rejector

Reject non-numeric, zero, or negative values for weight, height, reps, and durations (and negative calories or macros), with an error message naming the invalid field.

- For calories and macros, zero is accepted (a food can contain 0 g of fat).

3.1.16. FR16 – User data file format

Save each user's data automatically to its own JSON file.

- There is one file per profile, and the saving of data is automatic.
- Files are written safely so that an interrupted save does not damage the previous file (NFR6).

3.2. Non-functional Requirements

The system shall...

3.2.1. NFR1 (Constraint)

Run on Windows 10 and 11, be written in Java (or the final language choice), and use an existing JSON library for file I/O.

3.2.2. NFR2 (Performance)

Launch and show the profile selection screen within 10 seconds on a PC with 8 GB RAM and an SSD.

3.2.3. NFR3 (Performance)

Complete search, sort, and filter operations within 1 second, save a log entry within 3 seconds, and show the dashboard within 5 seconds for a profile with 2 years of logs.

3.2.4. NFR4 (Accuracy)

Keep the timers within ±1 second of real time for durations up to 20 minutes, with alerts firing within 1 second of zero.

3.2.5. NFR5 (Accuracy)

Calculate BMI to one decimal place, matching a reference calculator on all 20 test cases (imperial and metric), and keeping daily macro and calorie totals exact.

3.2.6. NFR6 (Reliability)

Leave the previous valid file intact if the app is killed during a save (write to a temp file, then rename).

3.2.7. NFR7 (Security / Privacy)

Make zero network connections and provide no encryption or access control (data is only as secure as the storage media; this replaces the old NFR4).

3.2.8. NFR8 (Usability)

Let at least 2 of 3 first-time testers create a profile and log a first workout in under 10 minutes without instructions.

3.2.9. NFR9 (Usability)

Name the offending field in 100% validation of error messages.

3.2.10. NFR10 (Scalability)

Support 20 profiles on one machine, each with up to 3 years of logs, while still meeting NFR2 and NFR3.

3.2.11. NFR11 (Maintainability)

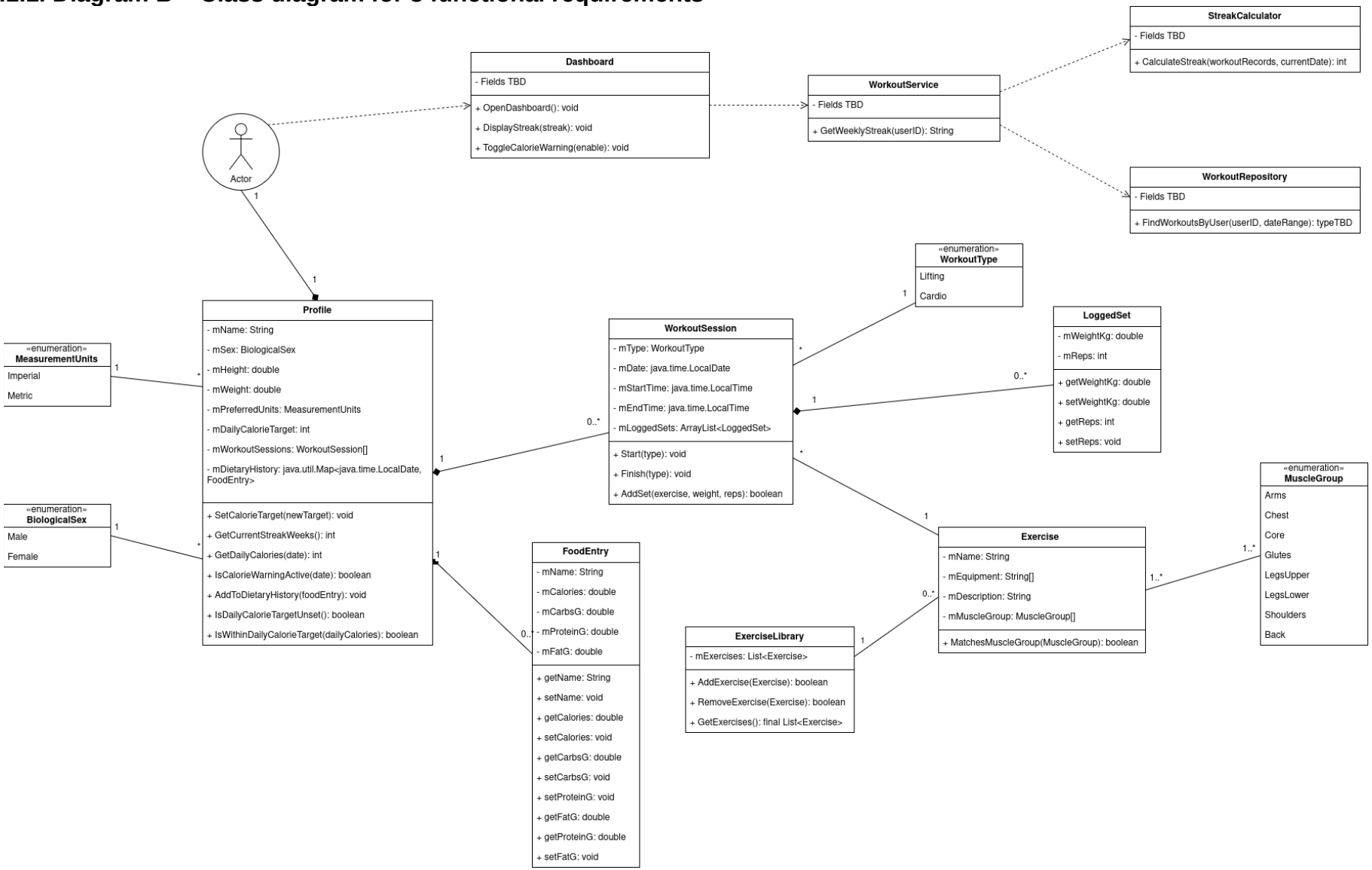
Have automated unit tests covering at least 80% of lines in the calculation modules (BMI, protein, calorie warning, streaks).

4. User Stories and Acceptance Criteria

4.1. Table A – User Stories

#	Source	User Story	Acceptance Criterion
US1	FR2	As a beginner who isn't sure where to start, I want to search for exercises by muscle group so that I can build a workout for the body area I want to train.	Given the exercise library is loaded, when I search for the muscle group "chest", every result shown is tagged "chest" and every chest-tagged exercise in the library appears.
US2	FR12	As an avid gym-goer, I want to see my streak of consecutive weeks with at least one workout so that I stay consistent while still taking the rest days my body needs.	Given I logged at least one workout in each of the last 3 calendar weeks (but none in the week before that), including a week with only one workout day, when I open the dashboard, then my streak displays "3 weeks."
US3	FR9	As a bodybuilder following a cutting or bulking plan, I want a warning when my logged calories are more than 20% off my daily target so that I can correct course for future days.	Given my daily calorie target is 2,000 kcal, when my logged total for the day reaches 2,401 kcal, then a warning is displayed (and none is displayed at exactly 2,400 kcal).
US4	NFR9	As a first-time user with no fitness-app experience, I want to set up my profile and log my first workout without reading instructions so that I don't get discouraged and quit.	Given three first-time testers with no instructions, when each is asked to create a profile and log one workout, then at least two of them finish within 10 minutes.

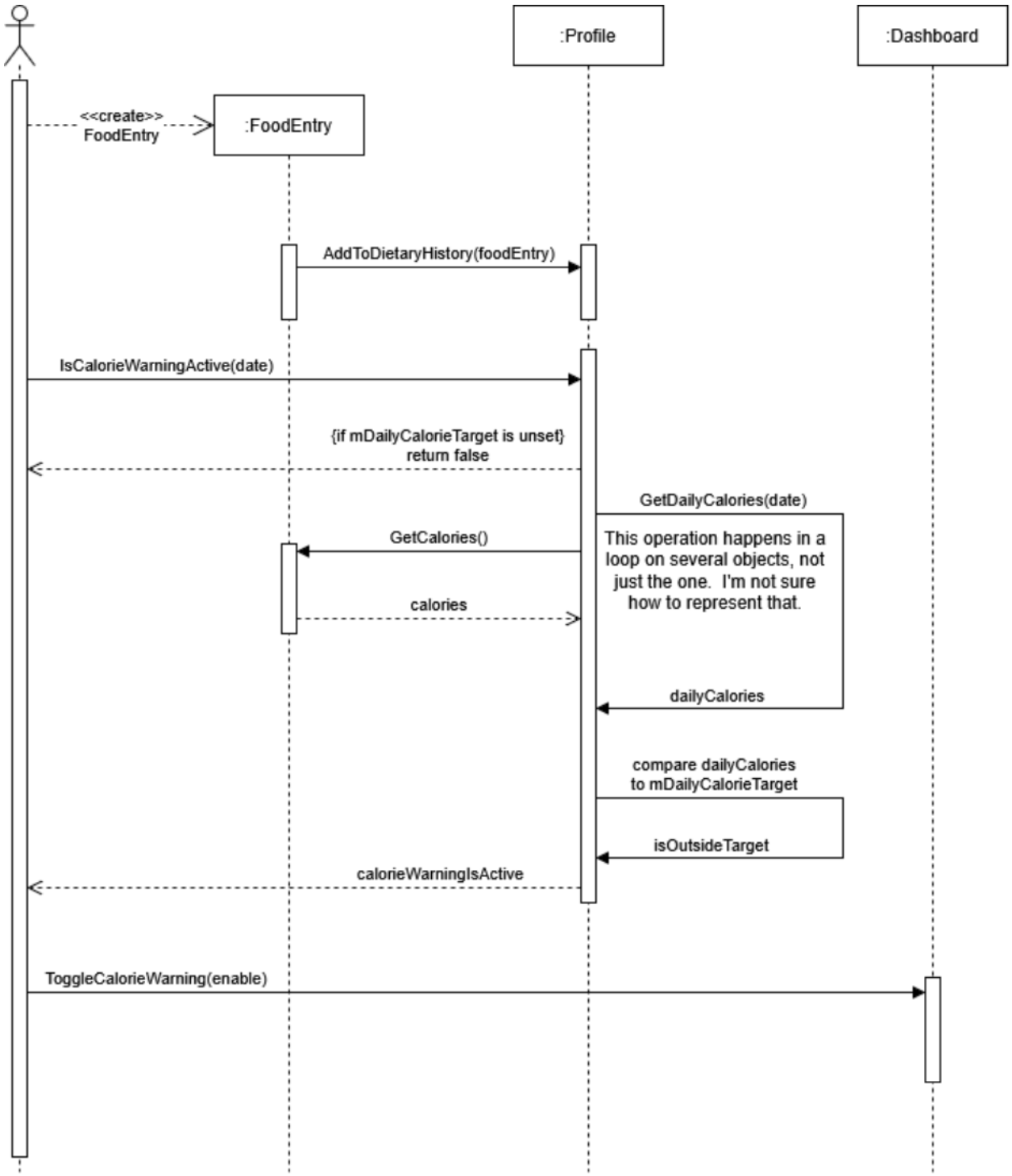
4.1.1. Diagram B – Class diagram for 3 functional requirements



4.2. Table B – Scenarios (for Bradley G)

Item	Text
User story	As a bodybuilder following a cutting or bulking plan, I want a warning when my logged calories are more than 20% off my daily target so that I can correct course for future days.
Acceptance criterion	Given my daily calorie target is 2,000 kcal, when my logged total for the day reaches 2,401 kcal, then a warning is displayed (and none is displayed at exactly 2,400 kcal).
Main flow	1. The user enters a food entry (name, calories, macros) for today. 2. A FoodEntry is created and its values are set (SetCalories, SetProtein, etc.) 3. The entry is submitted to the Profile **(through a yet-to-be-defined method)** which stores it in mDietaryHistory. 4. The caller automatically calls Profile.IsCalorieWarningActive(date) after the previous step so it may show a warning when appropriate. 5. Profile calls GetDailyCalories(date), which in turn FoodEntry.GetCalories() on each dietary history entry for that date to sum them up. 6. Profile compares the total to mDailyCalorieTarget and returns true if it is more than 120% or less than 80% of the target (false otherwise) 7. If Profile.IsCalorieWarningActive returns true, a warning is displayed **(through a yet-to-be-defined method)**.
Alternative flow	5a. If mDailyCalorieTarget is unset (let a value of 0 be the sentinel), then Profile.IsCalorieWarningActive returns early with the value false.

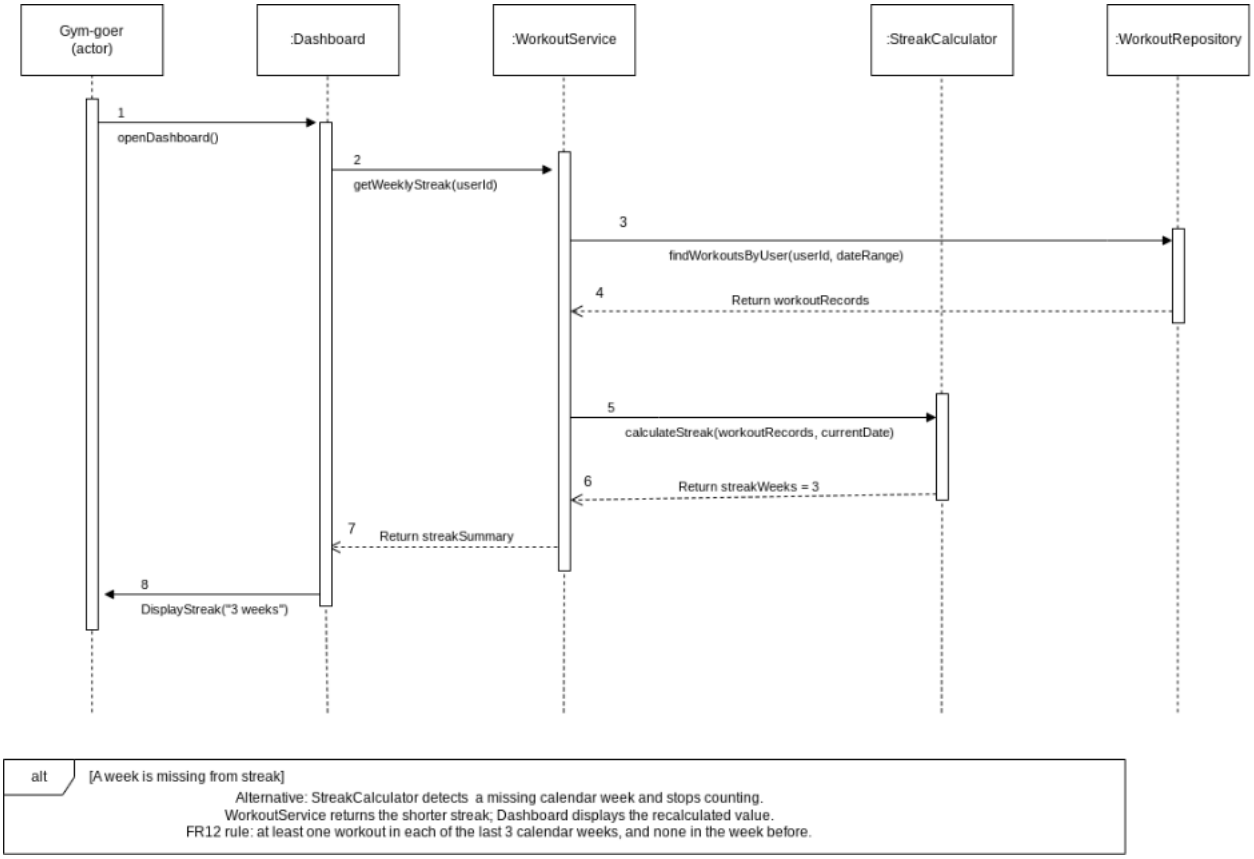
4.2.1. Diagram B – Sequence diagram for scenario 1



4.3. Table C – Scenarios (for Jakub N)

Item	Text
User story	As an avid gym-goer, I want to see my streak of consecutive weeks with at least one workout so that I stay consistent while still taking the rest days my body needs.
FR12 acceptance rule	The streak displays “3 weeks” when the user has logged at least one workout in each of the last three calendar weeks, including a week with only one workout day, and has logged no workout in the calendar week before those three weeks.
Main flow	1. The gym-goer opens the dashboard. 2. The dashboard requests the user's weekly workout streak from WorkoutService. 3. WorkoutService requests workout records for the relevant calendar weeks from WorkoutRepository. 4. WorkoutRepository returns the workout records. 5. WorkoutService asks StreakCalculator to calculate the consecutive weekly streak. 6. StreakCalculator checks that each of the last three calendar weeks contains at least one workout and that the week before them contains none. 7. StreakCalculator returns the calculated streak to WorkoutService, which sends the streak summary to the dashboard. 8. The dashboard displays “3 weeks” to the gym-goer.
Alternative flow	6a. If one of the required calendar weeks has no recorded workout, StreakCalculator stops counting at the first missing week. WorkoutService returns the shorter streak, and the dashboard displays the recalculated value instead of “3 weeks.”

4.3.1. Diagram A2 – Sequence diagram for scenario 2



5. Appendices

5.1. Glossary of terms

Term	Definition
BMI	Body mass index: a number calculated from a person's height and weight, grouped into categories such as underweight and normal.
Calander week	A week from Monday to Sunday
Class Diagram	A visual representation of the classes and relationships in software system.
Dashboard	The home screen of the app, summarizing the day's macros, calories versus target, and streak.
Exercise library	The built-in collection of exercises, each tagged with muscle groups and equipment.
Hydration reminder	A recurring reminder to drink water during an active workout session.
Macros	Macronutrients: carbohydrates, protein, and fat recorded in grams.
Muscle Group	A body region that exercises train: Arms, Chest, Core, Glutes, Legs Upper / Lower, Shoulders, or Back.
Profile	A user's stored details (name, sex, height, weight, units) and all of one’s workout and food data.
Reps	Repetitions: the number of times an exercise movement is performed in one set.
Rest Timer	A countdown between sets that alerts the user when the rest period is over
Sequence Diagram	A visual representation of the interactions between objects in a specific scenario.
Session	One workout, from the moment the user starts it until the user finishes it.
Set	A group of repetitions of one exercise, recorded with the weight.
Stakeholder	A person or group with an interest in the

	software system.
Streak	The number of consecutive calendar weeks that each contain at least one logged workout.
Use case	A description of a specific interaction between a user and the system.

5.2. List of acronyms and abbreviations

Acronym	Meaning
BMI	Body Mass Index
FR / NFR	Functional / Non-Functional Requirement
GUI	Graphical User Interface
JRE	Java Runtime Environment
JSON	JavaScript Object Notation
PC	Personal Computer
RAM	Random Access Memory
SRS	Software Requirements Specification
SSD	Solid State Drive
UC	Use Case
UML	Unified Modeling Language
US	User Story

5.3. References and Resources

CITATION 1 – Y-mart SRS Document

CITATION 2 – IEEE - The Design and Development of The Physical Fitness Test Management Information System. https://ieeexplore.ieee.org/document/9725086

CITATION 3 – Lift-off Fitness Tracker. https://liftoffrank.com/

CITATION 4 – Mass General Brigham. "How much protein when working out." Source of the protein ranges attributed to Wardlaw, who recommends that moderately active people aim for the lower end.https://www.massgeneralbrigham.org/en/health-wellness/how-much-protein-when-working-out